

DPEN021: Enabling Mathematics A

Class Questions Week 2

Week 2A

Key Idea - Simplifying Algebraic Fractions

1. Simplify the following:

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|--------------------------------|-------------------------------|--|
| (a) $\frac{2xy}{5yz}$ | (b) $\frac{m^2n}{5m^3n^2}$ | (c) $\frac{12x^4y}{8xy^2}$ |
| (d) $\frac{a}{a+a^3}$ | (e) $\frac{3x+x^2}{6x^2}$ | (f) $\frac{x-5}{7x-35}$ |
| (g) $\frac{5x+15}{7x+21}$ | (h) $\frac{a^2-2ab}{a-2ad}$ | (i) $\frac{5ax}{15a+10a^2}$ |
| (j) $\frac{3x^2}{15x^3+10x^2}$ | (k) $\frac{x^2-16}{x^2+4x}$ | (l) $\frac{7m^2-28m^2n}{16m-4m^2n}$ |
| (m) $\frac{x^2-49}{x^2-4x-21}$ | (n) $\frac{2z^3-54}{2z^2-6z}$ | (o) $\frac{y^2+7y+10}{y^2-25}$ |
| (p) $\frac{49-x^2}{x^2-49}$ | (q) $\frac{(x-2)^3}{x^2-4}$ | (r) $\frac{2x^2+20x+42}{2x^2+32x+126}$ |

Key Idea - Multiplication and Division of Algebraic Fractions

1. Simplify the following:

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| (a) $\frac{3}{7x^2} \times \frac{14x^4}{9}$ | (b) $\frac{\pi x^2}{4} \times \frac{A}{\pi x}$ | (c) $\frac{3ab}{5b^2} \times \frac{20a^4}{9a^3b}$ |
| (d) $\frac{3m-6}{4} \times \frac{8m}{m^2-2m}$ | (e) $\frac{x-1}{x+1} \times \frac{2x+4}{x^2+x-2}$ | (f) $\frac{15x^2-5xy}{10xy} \times \frac{2y}{3x-y}$ |
| (g) $\frac{2x+2}{x^2+4x+3} \times \frac{x+3}{7x+5}$ | (h) $\frac{a-b}{8} \times \frac{16}{b^2-a^2}$ | (i) $\frac{m^2-9}{m^2-m-12} \times \frac{m}{m^2-3m}$ |
| (j) $\frac{3mn}{10m} \div \frac{6n}{20m^2}$ | (k) $\frac{Q}{\pi d^2/4}$ | (l) $\frac{3x^2-6x}{15} \div \frac{5x-10}{5}$ |
| (m) $\frac{x^2-121}{x^2-4} \div \frac{x+11}{x+2}$ | (n) $\frac{3x-9}{x^2-4} \div \frac{x^2-5x+6}{x+2}$ | (o) $\frac{n^2-100}{5n-5p} \div \frac{(n-10)(n+10)}{n-p}$ |
| (p) $\frac{10}{a^2-a-20} \div \frac{4}{a^2-16}$ | (q) $\frac{5x+15}{x+1} \div \frac{x+3}{2x^2+x-1}$ | (r) $\frac{3x^3}{4a^2} \times \frac{ay-a}{xy^2} \div \frac{3y-3}{4ay^2}$ |

Key Idea - Addition and Subtraction of Algebraic Fractions

1. Simplify the following:

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|-------------------------------------|---------------------------------------|---|
| (a) $\frac{a}{3} + \frac{a}{7}$ | (b) $\frac{1}{x^2y} + \frac{2}{xy^2}$ | (c) $\frac{3b+1}{12} - \frac{b-2}{4}$ |
| (d) $\frac{1}{x+2} - \frac{1}{x+3}$ | (e) $\frac{1}{x+y} + \frac{1}{x-y}$ | (f) $\frac{1}{x-1} + \frac{5}{(x-1)^2}$ |

(g) $\frac{x^2}{x^2+3x+2} - \frac{2x}{x+2}$	(h) $x - y + \frac{x^2}{x+y}$	(i) $\frac{3}{b^2-4} - \frac{5}{3b-6}$
(j) $\frac{5}{3x} + \frac{2}{x^2-5x}$	(k) $\frac{3b}{a^2-b^2} + \frac{1}{a-b}$	(l) $\frac{3}{2k^2-4k} - \frac{1}{k-2}$
(m) $\frac{5}{2a+6} - \frac{a}{a^2-9}$	(n) $\frac{3}{x^2+2x-8} + \frac{2}{x^2+x-6}$	(o) $\frac{1}{x^2-4x+3} - \frac{1}{x^2-1}$
(p) $\frac{5x}{3} - \frac{2x+3}{4} + \frac{x}{6}$	(q) $\frac{x-3}{x-4} - \frac{x+4}{x+3} + \frac{1}{x-1}$	(r) $\frac{2a^2-b^2}{a^2} - \frac{b^2-c^2}{b^2} - \frac{c^2-a^2}{c^2}$

Week 2B

Key Idea - Rules for Surds

1. Simplify the following:

(a) $\sqrt{50} - \sqrt{32}$	(b) $\sqrt{a} - 3\sqrt{b} + \sqrt{4a}$	(c) $5\sqrt{54} + 2\sqrt{24}$
(d) $\sqrt{27} + \sqrt{108} - \sqrt{12}$	(e) $3\sqrt{32} - \frac{1}{5}\sqrt{50} + 8\sqrt{18}$	(f) $6\sqrt{18a} - 4\sqrt{50a} + 3\sqrt{72a}$
(g) $-4\sqrt{3} \times 2\sqrt{11}$	(h) $2\sqrt{5} \times 3\sqrt{15}$	(i) $3\sqrt{2} \times 5\sqrt{14} \times \sqrt{7}$
(j) $(3\sqrt{11})^2$	(k) $\sqrt{8a} \times \sqrt{18a}$	(l) $\frac{\sqrt{27}}{\sqrt{4}}$
(m) $\frac{6\sqrt{45}}{2\sqrt{5}}$	(n) $\frac{10\sqrt{10}}{15\sqrt{18}}$	(o) $\frac{4\sqrt{12x}}{8\sqrt{3x}}$

2. Expand and simplify the following:

(a) $4\sqrt{3}(\sqrt{3} + 2\sqrt{5})$	(b) $(\sqrt{7} + 2)^2$	(c) $(\sqrt{7} + 2\sqrt{5})^2$
(d) $(1 - \sqrt{3})(1 + \sqrt{3})$	(e) $(2\sqrt{6} + \sqrt{5})(2\sqrt{6} - \sqrt{5})$	(f) $(2\sqrt{7} - 5\sqrt{2})(2\sqrt{7} + 5\sqrt{2})$

Key Idea - Rationalising the Denominator

1. Express with a rational denominator:

(a) $\frac{1}{\sqrt{2}}$	(b) $\frac{2}{\sqrt{7}}$	(c) $\frac{1}{4\sqrt{5}}$
(d) $\frac{3}{2\sqrt{5}}$	(e) $\frac{\sqrt{7}}{\sqrt{3}}$	(f) $\frac{2\sqrt{3}}{\sqrt{5}}$
(g) $\frac{1}{\sqrt{3}-5}$	(h) $\frac{2}{\sqrt{6}-2}$	(i) $\frac{4}{\sqrt{7}-\sqrt{5}}$
(j) $\frac{\sqrt{3}}{3-\sqrt{5}}$	(k) $\frac{7}{\sqrt{6}-\sqrt{2}}$	(l) $\frac{\sqrt{3}}{\sqrt{3}+\sqrt{2}}$
(m) $\frac{2\sqrt{3}}{\sqrt{5}-\sqrt{2}}$	(n) $\frac{4\sqrt{3}-2\sqrt{2}}{7\sqrt{5}}$	(o) $\frac{3\sqrt{3}+\sqrt{2}}{2\sqrt{5}+3\sqrt{2}}$

Week 2C

Key Idea - Positive Index Rules

1. Simplify the following:

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| (a) $x^2 \times x^5$ | (b) $3x^4 \times 2x^3$ | (c) $a^7 \times a$ |
| (d) $a^5 \div a^4$ | (e) $x^8 \div x^3$ | (f) $\frac{12x^6}{3x^2}$ |
| (g) $(-4x^5) \times (3x^2)$ | (h) $6a^8 \div 2a^3$ | (i) $(7p^4) \times (-2p)$ |
| (j) $(3a^2) \times (-4a^3) \times (2a)$ | (k) $(2x^3) \times (4x^2) \div x$ | (l) $(-6p^6) \div 2p^3 \div p$ |
| (m) $x^3y^2 \times x^4y$ | (n) $x^5y^4 \div x^2y$ | (o) $3a^5b^2 \times 2ab^4$ |
| (p) $10p^6q^3 \div 5p^2q$ | (q) $(4x^2y^3) \times (-3xy)$ | (r) $(6x^5 \div 3x^2) \times x^4$ |
| (s) $(-8p^7 \div 4p^2) \div p^3$ | (t) $(2a^3b^2) \times (3a^4) \div ab$ | (u) $(-5x^4y^3 \times 2x^2) \div (5xy)$ |
| (v) $2^{5x} \times 8^x \div 4^{2x}$ | (w) $2^3 \times 4^2 \times 8^2$ | (x) $5^a \times 5^{-2a} \times 5^{4a}$ |

Key Idea - Positive Index Rules Continued

1. Simplify the following:

- | | | |
|------------------------------------|--|--|
| (a) $(ab)^2$ | (b) $(3a)^2$ | (c) $(2ab)^3$ |
| (d) $\left(\frac{x}{y}\right)^2$ | (e) $\left(\frac{2x}{y}\right)^2$ | (f) $\left(\frac{3a}{2b}\right)^2$ |
| (g) $(y^2)^3$ | (h) (a^5) | (i) $(ab^2)^3$ |
| (j) $\left(\frac{x^2}{y}\right)^3$ | (k) $\left(\frac{4p}{q^2}\right)^2$ | (l) $\left(\frac{-2a^2}{3b}\right)^2$ |
| (m) $(-2b^2)^2$ | (n) $(3x^2y)^3$ | (o) $(-4a^3b)^2 \times a$ |
| (p) $\frac{(-2x^2y)^3}{4x^3y^2}$ | (q) $\left(\frac{-3x^3}{2y}\right)^2 \div x$ | (r) $\frac{(3ab^2)^2 \times a^3}{-9a^4b}$ |
| (s) $((a^2)^3)^2$ | (t) $((x^2y)^3)^2$ | (u) $\left(\frac{-3x^2y}{2z}\right)^2 \times \frac{4z^2}{x}$ |

Week 2D

Key Idea – Zero Index Rule

1. Simplify and evaluate, where possible:

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| (a) $(ab)^0 + 6$ | (b) $5b^0 + 3^0$ | (c) $2\left(\frac{p}{q}\right)^0$ |
|------------------|------------------|-----------------------------------|

Key Idea – Negative Index Rules

1. Simplify the following expressions, express each answer with positive indices:

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|-----------------------------|-------------------------------------|--|
| (a) x^{-3} | (b) $(4b)^{-4}$ | (c) $5y^{-2}$ |
| (d) $\frac{1}{p^{-2}}$ | (e) $\frac{-2}{(3m)^{-3}}$ | (f) $\frac{3}{4a^{-4}}$ |
| (g) $\frac{x^{-2}}{y^{-3}}$ | (h) $\left(\frac{a}{b}\right)^{-2}$ | (i) $\left(\frac{2x^3}{y^2}\right)^{-1}$ |

2. Simplify the following expressions, express each answer with negative indices:

(a) $\frac{1}{x^4}$	(b) $\frac{3}{b^2}$	(c) $\frac{2}{3y^2}$
(d) $\frac{1}{x-2}$	(e) $\frac{2}{(4a-1)^3}$	(f) $\frac{1}{4(x+1)^6}$

Key Idea – Fractional Index Rule

1. Write the following in index notation.

(a) $\sqrt[4]{a}$	(b) $\sqrt[3]{x^2}$	(c) $(\sqrt[3]{x})^2$
(d) $\frac{1}{(\sqrt[4]{y})^8}$	(e) $\frac{2}{(\sqrt{k+1})^5}$	(f) $\frac{1}{4\sqrt[3]{(x-2)^2}}$
(g) $\sqrt[4]{\frac{x^3}{y}}$	(h) $\left(\frac{1}{\sqrt[3]{z^2}}\right)^5$	(i) $\frac{\sqrt[7]{k^4}}{\sqrt{s}}$

2. Write without fractional indices:

(a) $s^{\frac{1}{3}}$	(b) $3w^{\frac{1}{2}}$	(c) $(16r^4)^{\frac{1}{2}}$
(d) $k^{-\frac{1}{3}}$	(e) $(6y^3)^{-\frac{1}{5}}$	(f) $5(x-1)^{-\frac{5}{2}}$
(g) $\frac{(x+2)^{-\frac{2}{3}}}{3}$	(h) $\left(\frac{y^2}{z^5}\right)^{\frac{3}{2}}$	(i) $\left(\frac{1}{8t^4}\right)^{-\frac{2}{3}}$

Key Idea – Combined Index Rules

1. Write each expression in simplest index form, using positive indices where possible.

(a) $a^3b^{-2} \times (-2a^2b^2)^4$	(b) $(2x^2y)^2 \times x^3y^{-4}$	(c) $a^2b \times (b^2)^{-2} \times (a^3c^{-2})^3$
(d) $\frac{(-2xy)^2 \times 2(x^2y^{-1})^3}{8(xy)^{-3}}$	(e) $\frac{5x^5y^{-2} \times 3(xy^3)^2}{15x^{-2}y}$	(f) $\frac{(2m^2n)^3}{(mn^3)^2 \times (4m^2)^2}$
(g) $\frac{5^{2n} \times 5^{n-1}}{5^{2n+1}}$	(h) $\frac{3^{-2}x^{-1}}{3^{-3}x^{-2}}$	(i) $\frac{a^{m+n} \times b^{3m-n}}{a^{-n} \times b^{3n}}$
(j) $(a^{-1}b)^3 \times \left(\frac{1}{b^{-3}}\right)^{1/3}$	(k) $(y^{-\frac{1}{2}} + x^{-\frac{1}{2}})(y^{-\frac{1}{2}} - x^{-\frac{1}{2}})$	(l) $(x^{1/3})^2 \times (x^{-1}y^{-3})^{-1} \times (x^{-5/3})$
(m) $\sqrt{x^3} \times \sqrt[3]{x^2}$	(n) $\frac{a^2 \times \sqrt[3]{a}}{a^{2/3} \times \sqrt[4]{a}}$	(o) $\frac{(\sqrt[4]{a^3b^{-2}})^2}{a^{\frac{1}{2}}b^{-1}}$
(p) $\frac{(\sqrt[5]{a})^3 \times a^{-\frac{1}{2}}}{a^{\frac{4}{5}} \div \sqrt{a^{-3}}}$	(q) $\frac{\left(x^{\frac{2}{3}}y^{-1}\right)^2 \times x^{-\frac{5}{3}}}{y^{\frac{1}{2}}}$	(r) $\frac{(8m^{-2}n^4)^{\frac{1}{2}}}{\sqrt[3]{m^3n^{-6}}}$